
Bypassing PC1 Makes SAEs More Reproducible

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Abstract

Sparse autoencoders (SAEs) are widely used to decompose language model activations into interpretable features, but recent work finds that many features are not reproducible across random seeds. Intuitively, the more important a feature is, the more reproducible it should be. We test this hypothesis and find a surprising inversion: the 50 most important features by zero-ablation are among the least reproducible across seeds, far below dictionary-wide baselines. We trace this to residual stream anisotropy, a well-studied phenomenon in NLP embeddings that has been documented in transformer representations but not connected to SAE training. In transformer middle layers, a single principal component (PC1) explains 70–99.9% of activation variance, and its sparse decomposition is non-identifiable: different seeds learn different features that tile the same direction. These features dominate importance rankings, explaining the inversion. We propose an intervention: bypassing PC1 during SAE training, routing it around the SAE and restoring it during inference. Across 9 models from 5 families (124M–70B parameters), this intervention raises top-50 feature recovery to 74–97%. The recovered features show greater causal consistency under steering ($p < 0.01$ in 8 of 9 models), and shared features produce $4\times$ more accurate logit-lens predictions than orphan features. Our results reconcile recent SAE instability findings with the existence of dense model features: the dominant PC1 direction is real and reproducible, but the individual sparse coordinates used to reconstruct it are not. Once this component is separated, a reproducible core of sparse, semantically coherent SAE features emerges.

1 Introduction

Mechanistic interpretability aims to understand the internal computations of large language models by identifying the features they use. Sparse autoencoders (SAEs) have emerged as a leading tool toward this goal [Cunningham et al., 2023, Bricken et al., 2023]. An SAE is trained to reconstruct a model’s internal activations using a sparse set of learned directions, with the hope that each direction corresponds to a meaningful feature used by the model. If SAE features reflect real structure, then independently trained SAEs on the same model and data should tend to recover the same features.

Recent work calls this prediction into question. Paulo & Belrose [2025] trained SAEs with different random seeds on the same model and data and found that $\sim 70\%$ of features differ across seeds. This finding, alongside broader concerns about SAE utility [Kantamneni et al., 2025, Wu et al., 2025, Smith et al., 2025], has contributed to growing skepticism about whether SAEs recover meaningful model structure.

We revisit SAE reproducibility by asking a different question. Rather than asking whether SAE features *in general* are reproducible, we ask whether the features *that matter most* are reproducible. We measure each feature’s importance via zero-ablation: insert the SAE at a model layer, set one feature’s activation to zero, and record the increase in cross-entropy loss. We measure reproducibility as the maximum cosine similarity between each feature’s decoder direction and all directions in an independently trained SAE; a feature is “shared” if this exceeds 0.7, following Paulo & Belrose

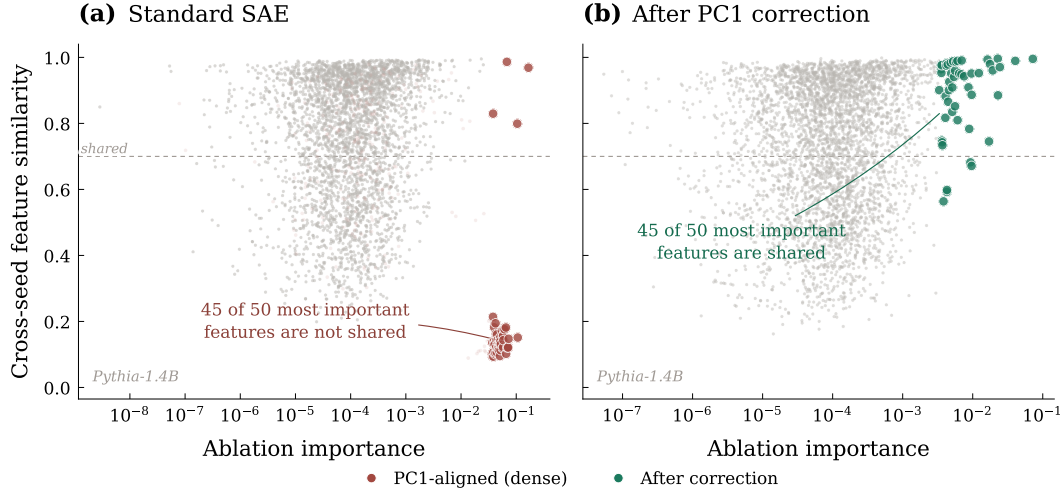


Figure 1: Each SAE feature plotted by its ablation importance (x -axis, log scale) and cross-seed decoder similarity (y -axis) on Pythia-1.4B. **(a)** Before PC1 correction: a small cluster of dense, PC1-aligned features (red) dominates the importance ranking despite near-zero reproducibility. **(b)** After PC1 correction: the cluster vanishes and the most important features become the most reproducible. The dashed line marks the shared/orphan threshold at cosine = 0.7.

[2025]. If importance reflects real model structure, the most important features should be especially likely to recur across seeds.

Surprisingly, the opposite is true: the most important features are the *least* reproducible. In most models we evaluate, the top-50 most important features are only 5–10% shared, far below dictionary-wide baselines of 44–62% (Figure 1, left).

We trace this inversion to a small population of features visible in Figure 1(a): a cluster in the bottom right that dominates the importance ranking despite near-zero cross-seed similarity. These features fire on nearly all tokens, and their decoder directions align with a single geometric structure: the first principal component (PC1) of the residual stream. This pattern is familiar from the NLP embeddings literature, where representations are known to cluster along a dominant direction — a phenomenon called anisotropy [Mu & Viswanath, 2018, Ethayarajh, 2019].

We propose that anisotropy is the root cause of the importance–reproducibility inversion. In transformer middle layers, PC1 explains 70–99.9% of activation variance, and the SAE must reconstruct this dominant signal. But no unique sparse decomposition of a continuous direction exists — different seeds can tile it with different near-equivalent feature sets. These seed-dependent features score highest on ablation importance because removing any one degrades the reconstruction of the dominant component, creating out-of-distribution magnitudes downstream.

The fix is correspondingly simple. We bypass PC1 during SAE training: the SAE is trained on activations with the PC1 projection removed, while PC1 is routed around the SAE and restored during inference. This removes the confound without discarding the dominant direction. After bypass, the top-50 recovery across seeds rises to 74–97%, and the relationship between importance and reproducibility reverses: the more important a feature, the more reproducible it is (Figure 1, right). These results are robust to the choice of preprocessing, importance metric, matching criterion, and SAE architecture (Section 5.3). Crucially, the downstream model still receives the full PC1 component; the bypass only prevents its arbitrary sparse decomposition from dominating feature-level analysis.

We also show that the recovered features are not only geometrically stable but causally meaningful: cross-seed decoder similarity predicts consistency under steering interventions, and shared features better predict their effects on output logits than orphan features.

Our work gives a specific diagnosis of SAE instability. The issue is not that all high-importance SAE features are arbitrary, but that a standard importance metric is dominated by a non-identifiable dense

71 component. Separating this component changes which features are selected as important, and the
72 resulting features are much more stable across training runs.

73 2 Background

74 **SAEs and the superposition hypothesis.** Sparse autoencoders decompose a model’s internal
75 activations into sparse combinations of learned directions [Cunningham et al., 2023, Bricken et al.,
76 2023]. Gao et al. [2024] scaled the approach to GPT-4 and introduced the TopK activation function;
77 Templeton et al. [2024] applied SAEs to Claude 3 Sonnet. These successes led to optimism that
78 SAEs could, in principle, fully enumerate the features a model uses, enabling safety auditing and
79 circuit-level understanding.

80 The theoretical foundation is the superposition hypothesis [Elhage et al., 2022]: in toy models, more
81 important features receive dedicated linear directions, whereas less important ones are compressed
82 into shared subspaces where they interfere. An SAE, by definition, searches for sparse linear
83 combinations that reconstruct the activation. It should therefore recover dedicated directions most
84 reliably, since they are the strongest, most isolated signals in the representation — exactly the kind of
85 structure that sparse coding is designed to find. This yields the prediction we test: more important
86 features should be more reproducible across independently trained SAEs.

87 **Residual stream anisotropy.** Contextual representations in transformers occupy a narrow region
88 of the activation space rather than spreading uniformly. Ethayarajh [2019] showed that BERT, ELMo,
89 and GPT-2 representations cluster in a narrow cone, with average cosine similarity increasing in
90 deeper layers. Mu & Viswanath [2018] proposed removing the mean and top principal components
91 (“All-but-the-Top”), which restored isotropy and improved downstream performance. Razzhigaev
92 et al. [2024] extended this literature to modern architectures, documenting a bell-shaped anisotropy
93 profile in decoder-only transformers: low at the embedding layer, peaking in the middle, and declining
94 toward the output.

95 Prior work has established that this anisotropy is specifically rank-1; i.e., dominated by a single
96 direction rather than a broader low-rank subspace: gradient drift from cross-entropy training produces
97 a rank-1 mean shift in embeddings [Gao et al., 2019, Biś et al., 2021]; self-attention contracts token
98 representations doubly exponentially toward a single leading eigenvector [Dong et al., 2021, Noci
99 et al., 2022]; and the additive structure of the residual stream compounds any consistent directional
100 component across layers [Godey et al., 2024].

101 **Dense SAE features.** Sun et al. [2025] identified “dense latents” — SAE features that fire on nearly
102 all tokens — and showed they often reflect real model structure. They also observed that dense latents
103 are not identifiable across seeds: different SAEs learn different bases for the same shared subspace,
104 because linear combinations reconstruct equally well. Our work builds on this observation. Sun et
105 al. also retrained SAEs after ablating the dense subspace to show these features are intrinsic; our
106 bypass is a similar intervention, but aimed at resolving the PC1 confound rather than characterizing
107 dense latents. We identify the subspace geometrically, via decoder alignment with PC1, rather than
108 by firing frequency — and show that frequency-based identification is insufficient (Section 5.3). We
109 also connect the subspace to residual stream anisotropy and show that PC1 projection is primarily
110 associated with activation magnitude (Appendix A).

111 3 The PC1 Confound

112 We now describe how these two phenomena (residual stream anisotropy and dense latent non-
113 identifiability) combine to produce the importance–reproducibility inversion. We present supporting
114 evidence for each component of this account and derive testable predictions in Section 3.4.

115 3.1 PC1 in the residual stream

116 Across the 9 models we evaluate, PC1 of the centered residual stream at the middle layer explains
117 70–99.9% of the total activation variance, exceeding 93% in 7 of 9 models. This dominance follows
118 a bell-shaped profile across layers: on Pythia-160M, PC1 explains 5.1% at the embedding, peaks

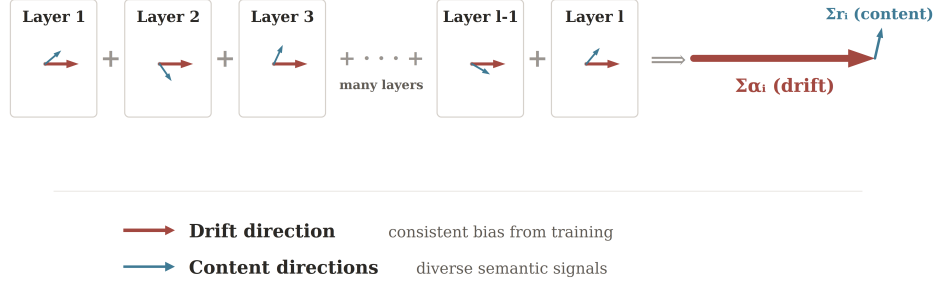


Figure 2: Residual stream accumulation produces rank-1 dominance. Each layer contributes drift along a shared direction (red) and content orthogonal to it (blue). The consistency of the drift direction is driven by gradient drift, attention contraction, and skip-connection compounding (Section 2). After L layers, the drift direction carries 93–99.9% of total variance.

at 89.2% at layer 5, and drops to 7.5% at the final layer (Appendix A). An untrained Pythia-160M shows PC1 $\approx 5\%$ throughout, implying that training creates the anisotropy.

PC1 aligns with the mean activation direction ($\cos = 0.92$ on Pythia-1.4B) and explains 99.9% of the variance in activation norm ($R^2 = 0.999$): it primarily tracks activation magnitude rather than token identity or semantic content. The dominance is sharply rank-1 — PC2 explains only 0.7% at the peak layer (Appendix A) — consistent with a spectral gap exceeding $50,000\times$ between the first and second eigenvalues (Appendix L). The mechanism is visualized in Figure 2.

3.2 Why the SAE’s decomposition of PC1 is non-identifiable

The SAE must reconstruct its input, and PC1 carries $>93\%$ of the input variance. With sparsity $k = 64$ and an overcomplete dictionary of $4d$ directions in a d -dimensional space, the reconstruction of a dominant dense direction need not have a unique sparse representation. This is standard in sparse coding: when the target signal is not itself sparse, multiple sparse codes can achieve comparable reconstruction error [Donoho & Elad, 2003]. PC1 is precisely such a signal — a single continuous direction shared by all tokens, with no sparse feature structure for the SAE to discover. The SAE therefore has many near-equivalent ways to allocate reconstruction of $\mathbf{v}_1$ across dictionary elements, and different initializations can converge on different allocations (Figure 3).

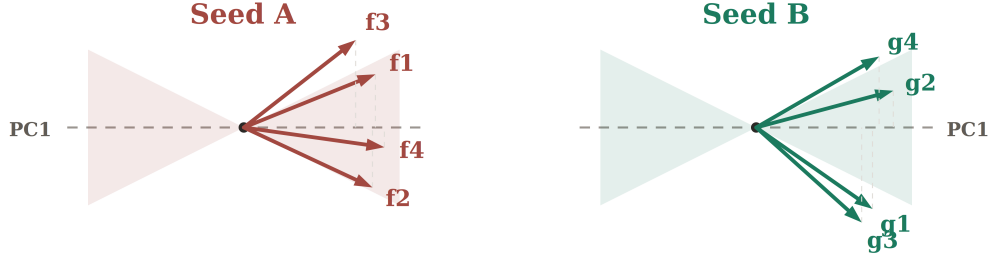
If several decoder atoms point near $\mathbf{v}_1$, any partition of the PC1 score $a_t = \langle \mathbf{x}_t - \boldsymbol{\mu}, \mathbf{v}_1 \rangle$ across those atoms achieves nearly identical reconstruction loss. TopK exacerbates this because all k selected features receive reconstruction gradients with no ℓ_1 penalty or diversity term to consolidate the signal into fewer atoms. The result is a large basin of near-equivalent PC1 tilings. Different initializations converge to different co-firing, PC1-aligned feature sets.

Empirically, five of the top six features by importance in Pythia-1.4B fire on the exact same 773 token positions (Jaccard overlap 1.000) with activation magnitudes correlated at $r > 0.94$. Two random features at the same firing rate would share approximately 12 positions. These five features are reconstructing one signal five ways.

The subspace is nonetheless recovered: dense features from two independently trained SAEs on Pythia-1.4B span overlapping subspaces (principal angle overlap 0.758; Appendix B). PCA recovers the same subspace with overlap 0.943. Both SAEs find the same structure; the sparsity constraint forces them to express it in different, seed-dependent coordinates.

3.3 Why dense features dominate importance rankings

Zero-ablation — setting one feature’s activation to zero and measuring the loss increase — is the standard metric for feature importance. For sparse features, the loss increase measures the removal of a particular signal the model relies on. For dense features, removing one destroys part of the SAE’s reconstruction of PC1, creating out-of-distribution activation magnitudes downstream. The importance score reflects the SAE’s dependence on its own decomposition of PC1 — it measures how much the reconstruction degrades, regardless of whether the feature contributes to the model’s computation.



Subspace shared (overlap 0.758) but individual features are not (cosine < 0.7).

Figure 3: Non-identifiability of sparse tiling. Two SAEs decompose PC1 using different dictionary vectors (red vs. green). Both reconstruct equally well, but individual features have low cross-seed cosine similarity.

Group ablation confirms redundancy: in Pythia-1.4B, the importance of individual dense features sums to $2.18\times$ the joint effect of ablating all 502 simultaneously (Appendix C). This inflation crowds out semantic features in the importance ranking.

3.4 Hypothesis and predictions

The observations above yield four testable predictions:

1. *Top-50 inversion.* In models where PC1 is dominant, the 50 most important features should be disproportionately *unshared* across seeds, because dense features dominate the importance ranking and are non-identifiable.
2. *PC1 bypass reversal.* Removing PC1 before SAE training should reverse the relationship: without the confound, importance should predict reproducibility.
3. *Scaling with R .* The bypass lift should scale with $R = PC1/(1 - PC1)$, the ratio of variance captured by the confounding direction to the variance available for content features.
4. *Causal faithfulness.* After bypass, the recovered features should be not only geometrically stable but causally faithful: their decoder directions should better predict their effects on model outputs, and matched features from independent SAEs should steer the model in similar directions.

We test Predictions 1–3 in Section 5 and Prediction 4 in Section 6.

4 Experimental Setup

Models. We evaluate on 9 models from 5 families: Pythia-160M, 1.4B, and 6.9B; GPT-2-small; Gemma 2 2B and 9B; Llama 3.1 8B and 70B; and Qwen 2.5 7B. All experiments target the middle residual-stream layer of each model (Table 7). Activations are extracted from the output of the transformer block at the target layer (post-attention, post-MLP, pre-normalization of the next layer).

SAE architecture and training. We train TopK sparse autoencoders following Gao et al. [2024], with sparsity $k = 64$ and dictionary size $4\times$ the model’s hidden dimension.¹ The encoder and decoder are linear layers with tied initialization; decoder columns are unit-normalized after each gradient step. We train on 5M tokens from WikiText-103 for 20,000 steps with batch size 4096 and learning rate 3×10^{-4} (Adam). For each model and condition, we train five SAEs with independent random seeds (42, 137, 0, 1, 2), except Llama 3.1 70B, where we train two seeds (42, 137) for compute reasons. Full hyperparameters are in Appendix E.

¹Our main experiments train on centered pre-normalization residual stream activations. Gao et al. [2024] additionally normalize inputs to unit norm, which removes scale information. We test this and three other preprocessing variants in Section 5.3 and Appendix M: the confound persists under unit-norm normalization on Pythia-1.4B (top-50 sharing 4%), and PC1 bypass provides improvement across almost all settings.

Table 1: Top-50 sharing rates before and after PC1 bypass across 9 models. Dict-wide: baseline fraction of all alive features shared (max cosine > 0.7) with another seed. Top-50: sharing rate among the 50 most important features. Values for five-seed models are mean $\pm$ std across seed pairs involving the reference seed; Llama 70B uses two seeds.

Model	Family	PC1 (%)	Dict-wide (%)	Top-50 (%)	Top-50 bypass (%)	Lift
Qwen 2.5 7B	Qwen	99.9	44 ± 1	8 ± 0	97 ± 1	+89
Llama 3.1 70B	Llama	97.8	56	38	84	+46
GPT-2-small	GPT-2	97.0	45 ± 1	6 ± 0	74 ± 3	+68
Llama 3.1 8B	Llama	96.9	61 ± 1	46 ± 0	95 ± 2	+49
Pythia-1.4B	Pythia	96.4	62 ± 0	10 ± 0	90 ± 1	+80
Pythia-6.9B	Pythia	95.7	62 ± 3	6 ± 0	92 ± 3	+86
Pythia-160M	Pythia	93.4	49 ± 0	5 ± 1	79 ± 3	+74
Gemma 2 2B	Gemma	78.2	60 ± 1	75 ± 3	91 ± 2	+16
Gemma 2 9B	Gemma	69.9	59 ± 1	87 ± 1	91 ± 1	+4

PC1 estimation and bypass. We estimate PC1 from 50,000 centered activations sampled from the same distribution used for SAE training. In the bypass condition, we subtract the PC1 projection from centered activations before passing them to the SAE:

$$\mathbf{x}_{\text{bypass}} = (\mathbf{x} - \boldsymbol{\mu}) - ((\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{v}_1) \mathbf{v}_1 \quad (1)$$

During inference, the PC1 component is routed around the SAE and added back before the reconstructed activation continues to the next layer. The downstream model therefore receives the original PC1 component together with an SAE reconstruction of the remaining residual-stream component.

Importance. We measure each feature’s causal importance via zero-ablation: we insert the trained SAE at the target layer, set one feature’s activation to zero, and record the increase in cross-entropy loss over 256 held-out sequences from WikiText-103 validation. Features with higher loss increase are ranked as more important.

Reproducibility. We measure cross-seed reproducibility by computing the maximum cosine similarity between each feature’s decoder direction in one SAE and all decoder directions in another independently trained SAE, following Braun et al. [2024]. We restrict dictionary-wide statistics to *alive* features, defined as features that activate at least once on the held-out evaluation tokens. A feature is *shared* if its maximum cross-seed decoder cosine exceeds 0.7 [Paulo & Belrose, 2025]; otherwise it is an *orphan*. Results are robust to thresholds from 0.5 to 0.95 (Appendix F).

Main statistic. Our primary metric is *top-50 recovery*: the fraction of the 50 most important features (by zero-ablation) in one SAE that are shared with another independently trained SAE. For five-seed models, we report the mean $\pm$ standard deviation across the four seed pairs involving the reference seed (seed 42). This metric tests whether importance selects for reproducible structure or for seed-dependent artifacts.

Code for all experiments, including SAE training, evaluation, and analysis scripts, is included as supplementary material.

5 Results

5.1 The importance–reproducibility inversion

We rank features by ablation importance and check whether the most important features are shared across seeds. Table 1 reports top-50 sharing rates for all 9 models under both conditions, ordered by PC1 dominance.

In most highly-dominated PC1 models, the top-50 most important features are only 5–10% shared, far below dictionary-wide baselines of 44–62% (Prediction 1). Llama is a partial exception. In Gemma 2 9B, where PC1 explains only 70%, the pattern reverses: top-50 sharing is 87%, above baseline.

These confounding features constitute 2–16% of the alive dictionary but capture 49–90% of total ablation importance (Appendix D). Table 2 illustrates the pattern on Qwen 2.5 7B: the top three

Table 2: Top features by importance before PC1 bypass in Qwen 2.5 7B. Confounding features (top) fire on nearly all tokens with identical distributions. The shared features (bottom) fire on specific token classes and have near-zero PC1 alignment.

Feature	Imp.	Firing	PC1 $\cos^2$	Shared	Top tokens
14126	0.537	98.2%	0.260	×	<i>non-selective (fires on all tokens)</i>
9091	0.522	98.2%	0.229	×	
9651	0.462	98.2%	0.243	×	
516	0.188	3.0%	0.001	✓	spaces before numbers
3974	0.112	5.7%	0.003	✓	<i>The, the</i>
2094	0.061	3.6%	0.000	✓	sentence-ending periods

Table 3: Top features by importance after PC1 bypass (Pythia-1.4B). Each feature fires on a coherent semantic or syntactic class and is recovered at high cosine similarity across seeds.

Feature	Cosine	Freq	Example contexts
5748	0.989	4.0%	“off the coast of Mexico”; “Numerous shipwrecks ”; “the seas around Papa St”
3415	0.961	2.2%	“completed a flying instructors course”; “No. 2 Air Navigation School”; “There is an airstrip ”
7668	0.971	0.2%	“Slayer have headlined ”; “is headquartered at”; “(7), headquartered at”

features by importance fire on 98.2% of tokens with identical distributions and no cross-seed match, while the only shared features in the top 50 fire on narrow, identifiable token classes with near-zero PC1 alignment.

Bypassing PC1 during SAE training reverses the relationship (Prediction 2). On Pythia-1.4B, top-50 sharing rises from 10% to 90%, and the cross-entropy overhead improves slightly (3.197 to 3.186). Removing further principal components adds nothing; top-50 sharing is flat from 1 to 50 PCs removed (Appendix G). Figure 1 shows the effect on the full feature distribution: the dense cluster that dominates the importance ranking in (a) vanishes in (b), and the most important features become the most reproducible.

Table 3 shows the most important features after bypass on Pythia-1.4B. Where the confounding features fired on everything and differed across seeds, these features fire on narrow, coherent token classes and are recovered at cosine >0.95 by an independently trained SAE. Of the top 50 after bypass, 45 are shared (90%). A post-hoc alternative that excludes dense features without retraining achieves 96% (Appendix H).

5.2 Scaling with R

The bypass lift should scale with $R = \text{PC1}/(1 - \text{PC1})$, the ratio of variance captured by the confounding direction to the variance available for content features (Prediction 3). Table 1 supports this across 9 models from 5 families. Figure 4 plots the lift against R on a log scale (Pearson $r = 0.73$ on $\log R$, $p = 0.026$). At Qwen 2.5 7B ($R \approx 999$), the confound is total: zero overlap between pre- and post-bypass top-50 features. At Gemma 2 9B ($R \approx 2.3$), the top features are already semantic and the bypass adds only 4 points. Both Llama models show lower lift than predicted, suggesting their residual stream organizes content information differently despite similar PC1 dominance.

5.3 Controls

On an untrained Pythia-160M with random weights, PC1 explains only 4.7% of variance and the importance–reproducibility inversion is absent (top-50 sharing = 20% against a 3% baseline), indicating the confound requires learned anisotropy. To test whether removing any set of outlier features would produce the same effect, we exclude 679 features at random — matching the count removed by the PC1 criterion — across 1,000 trials. The mean top-50 sharing is 9% and the maximum is 12%; the PC1 criterion achieves 96%. The improvement is specific to the geometric criterion, not a consequence of excluding outliers in general.

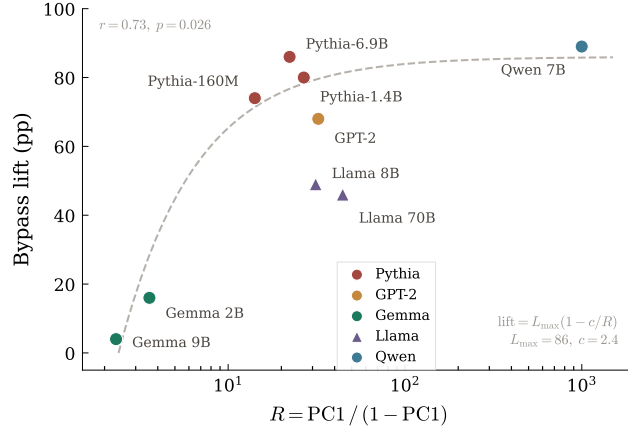


Figure 4: Bypass lift versus $R = \text{PC1} / (1 - \text{PC1})$. The dashed curve ($\text{lift} = 86(1 - 2.4/R)$) shows that lift saturates with increasing PC1 dominance. Llama models (triangles) are partial outliers excluded from the fit. Pearson $r = 0.73$ on $\log R$ across all 9 models ($p = 0.026$, including Llama).

248 The effect holds across layers 6, 12, and 18 of Pythia-1.4B (Appendix I), across domains when training
 249 on OpenWebText instead of WikiText (Appendix J), and across top- k from 20 to 500 (Appendix K).

250 A natural alternative to geometric identification is to exclude features by firing frequency, as proposed
 251 by Sun et al. [2025]. Excluding features with firing rates above 3%, 5%, 10%, or 15% produces top-50
 252 sharing of 0–4% — no better than no exclusion. This fails because many confounding features fire
 253 on only 1–5% of tokens individually while aligning collectively with PC1; geometric identification is
 254 necessary.

255 We further test whether the inversion depends on our preprocessing choices, importance metric, or
 256 matching method. On Pythia-1.4B, unit-norm normalization of SAE inputs — the preprocessing
 257 used by Gao et al. [2024] — does not resolve the confound: top-50 sharing is 4%, lower than
 258 without normalization. PC1 bypass raises it to 70%. The inversion also persists under mean-ablation
 259 importance (6% top-50 sharing without bypass, 90% with), confirming it is not specific to zero-
 260 ablation. Results are unchanged under mutual-nearest-neighbor and Hungarian one-to-one matching.
 261 The confound also extends beyond TopK: BatchTopK SAEs show the same pattern (2% $\rightarrow$ 92%
 262 on Pythia-1.4B). On GPT-2, unit-norm partially mitigates the inversion (10% $\rightarrow$ 36%) but does
 263 not eliminate it; PC1 bypass provides additional improvement (36% $\rightarrow$ 62%). Full results are in
 264 Appendix M.

265 6 Causal Validation

266 Predictions 1–3 established that PC1 bypass produces geometrically reproducible features. Predic-
 267 tion 4 asks whether these geometrically reproducible features also have more consistent causal effects
 268 across seeds. Following Templeton et al. [2024], we evaluate this via activation steering and logit
 269 lens analysis.

270 **Steering.** We inject matched feature decoder directions from two independently trained SAEs into
 271 the residual stream during generation ($\alpha = 5$) and measure the KL divergence between the resulting
 272 output distributions. On Pythia-1.4B, shared features produce cross-seed steering divergence of
 273 0.001, which is $6\times$ smaller than the steering effect itself (0.006) — both SAEs steer the model to the
 274 same place. For orphan features, cross-seed divergence (0.010) is $2\times$ larger than the steering effect
 275 (0.005); the disagreement between SAEs exceeds the effect of steering. Table 4 shows representative
 276 examples: matched features from seeds 42 and 137 steer the model toward the same semantic domain
 277 despite being trained from different initializations.

278 **Cross-seed steering consistency.** For each model, we sample 60 PC1-corrected features spanning
 279 the importance distribution, steer with the matched decoder direction from two independently trained
 280 SAEs, and measure the KL divergence between the two steered output distributions across six prompts.
 281 In 8 of 9 models, higher decoder similarity predicts lower steering divergence (Spearman $\rho = -0.35$

Table 4: Steering with shared features from two independently trained SAEs on Pythia-1.4B ($\alpha = 5-10$). Both SAEs steer toward the same semantic domain despite different random seeds.

Feat.	Cos.	Prompt	Unsteered	Seed 42	Seed 137
5748	.989	<i>The city was known for its</i>	many parks and gardens, and the city’s main park, the Central Park...	many canals , used for shipping . Also known for its many shipyards ...	many shipyards , also known for its many canals . The city’s fish-ing industry...
3415	.961	<i>In the summer of 1987,</i>	the first of the two-year-old buildings was opened to the public...	the first of the two-year-old F-15s was delivered to the United States Air Force ...	the first of the two-year-old Boeing 747-400s was delivered to the airline ...

to -0.66 , all $p < 0.01$; Appendix N). Gemma 2 9B is the exception, likely because its low PC1 dominance means most features are already well-matched.

Logit lens faithfulness. Each feature’s decoder direction, projected through the unembedding, predicts which tokens the feature should promote. We clamp each feature at $+5\sigma$ and measure whether the output distribution shifts toward these predicted tokens. Shared features produce a $4.22 \times$ probability increase toward their predicted tokens. Orphan features produce $1.20 \times$ ($p < 10^{-10}$, Mann-Whitney U). The decoder directions of shared features predict their causal behavior; orphan features’ decoder directions do not.

7 Discussion and Conclusion

Our results offer a specific diagnosis of SAE instability: a single dominant direction in the residual stream produces non-identifiable dense features that confound zero-ablation importance rankings. Bypassing this direction during SAE training resolves the confound across 9 models, and the resulting features are both reproducible and causally consistent. We discuss the implications and limitations of this finding.

Importance metrics. The inversion holds under zero-ablation and mean-ablation importance (Appendix M); whether it extends to gradient-based attribution or activation patching remains untested.

PC1 and dense latents. PC1 is highly correlated with activation magnitude ($R^2 = 0.999$) and moderately with token frequency ($\rho = 0.53$); it may carry functionally relevant information about entropy, confidence, or routing. The PC1-aligned features we identify correspond to one category in the dense latent taxonomy of Sun et al. [2025], who also document position tracking, context binding, and entropy regulation. Our results do not imply that all dense latents are confounds. What they share is that dense, continuous signals lack the sparse structure that makes dictionary learning well-posed — any SAE trained on them will produce non-identifiable features. Tools designed for continuous directions (PCA, linear probes, supervised steering vectors) are better suited to studying these components. Clarifying which dense latents are identifiable, which are functionally important, and which confound standard metrics remains open.

Scope. Our main experiments use TopK SAEs, but the confound also appears in BatchTopK (Appendix M). Our evaluation covers English-language text (WikiText-103 and OpenWebText); the effect on code, multilingual, or instruction-tuned settings is unknown. Our causal validation uses steering and logit lens primarily on Pythia-1.4B; broader evaluation across models and tasks would strengthen the result.

Conclusion. The instability reported in recent work is real, but our results suggest that much of it is concentrated in a geometrically identifiable confound. Practitioners training or evaluating SAEs on middle-layer residual stream activations should account for PC1 dominance, either by bypassing it during training or excluding PC1-aligned features post-hoc. Once this is done, ablation importance becomes a better predictor of cross-seed recovery: the most important features are the most reproducible.

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A PC1 Characterization

Methodology. We collect 5M token activations from the residual stream at layer 12 of Pythia-1.4B, using WikiText-103. Activations are extracted from the output of the transformer block (post-attention, post-MLP, pre-LayerNorm of the next layer). We center the activations by subtracting the dataset mean $\mu = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$. We then compute PCA on a subsample of 50,000 centered activations cast to float64: we form the covariance matrix $\mathbf{C} = \frac{1}{N-1} \mathbf{X}^\top \mathbf{X}$ and eigendecompose it via `np.linalg.eigh`. PC1 is the eigenvector corresponding to the largest eigenvalue. PC1 variance explained is $\lambda_1 / \sum_i \lambda_i$.

PC1 is primarily associated with activation magnitude. We test whether PC1 corresponds to a semantic direction or to activation magnitude. Three analyses indicate the latter:

Alignment with the mean. If PC1 arises from accumulated gradient drift (Section 2), it should align with the mean activation direction. We compute the cosine similarity between PC1 and the (uncentered) mean activation vector $\bar{\mathbf{x}}$. On Pythia-1.4B, $\cos(\text{PC1}, \bar{\mathbf{x}}) = 0.92$, confirming that PC1 is approximately the direction of the dataset mean—consistent with the rank-1 drift mechanism of Biś et al. [2021].

Activation norm. If PC1 captures the dominant source of activation magnitude, its projection should predict activation norm. For each activation $\mathbf{x}_i$, we compute the PC1 projection $p_i = (\mathbf{x}_i - \mu)^\top \mathbf{v}_1$ and the activation norm $\|\mathbf{x}_i\|_2$. We find Pearson $r = -0.9997$ ($p \approx 0$), giving $R^2 = 0.999$. PC1 projection explains 99.9% of the variance in activation norm. The negative sign reflects the arbitrary sign convention of eigenvectors. Tokens vary along PC1 primarily in magnitude, not in identity or meaning.

Token frequency. We compute the Spearman rank correlation between $|p_i|$ and the corpus frequency of the corresponding token. We find $\rho = 0.53$ ($p \approx 0$): more frequent tokens tend to have larger PC1 projections, consistent with the Zipfian frequency bias described by Gao et al. [2019].

These results suggest that PC1 primarily tracks activation magnitude rather than semantic content. This is consistent with the non-identifiability described in Section 3.2: a direction encoding scale has no sparse structure for the SAE to discover, so any tiling works equally well.

Bell-shaped layer profile. The theoretical account in Appendix L predicts that PC1 dominance should increase with depth as drift accumulates across layers, but decrease near the output, where the model must produce content-discriminative representations for next-token prediction via the unembedding matrix.

We verify this on Pythia-160M (12 layers, $d = 768$). For each layer, we collect 500,000 activations from WikiText-103, center them, and compute PC1 variance explained as above. We repeat the procedure on an untrained Pythia-160M initialized with random weights (same architecture, no training) to confirm that training creates the anisotropy.

Table 5 reports the results. In the trained model, PC1 variance explained rises from 5.1% at the embedding layer to a peak of 89.2% at layer 5, then declines to 7.5% at the final layer — a clear bell-shaped profile. The spectral gap is sharp: PC2 explains only 0.7% at the peak layer, confirming rank-1 rather than low-rank structure. In the untrained model, PC1 remains at 4–5% throughout — far below the trained model’s peak of 89.2%, confirming that training creates the anisotropy.

B Dense Subspace Recovery

We test whether independently trained SAEs recover the same dense subspace, even though individual dense features differ across seeds.

Method. We identify dense features in each SAE as those whose decoder direction has $\cos^2$ with PC1 exceeding $5\times$ the median across alive features. For two SAEs trained with seeds 42 and 137 on Pythia-1.4B (layer 12), this yields 502 and 487 dense features respectively. We extract the corresponding decoder directions (unit-normalized rows of the decoder weight matrix) and compute the principal angles between the two subspaces spanned by these directions.

Table 5: PC1 and PC2 variance explained (%) across layers of Pythia-160M.

Layer	Trained		Untrained	
	PC1	PC2	PC1	PC2
Emb	5.1	4.4	5.1	4.2
L1	10.1	4.4	5.0	4.8
L2	14.4	5.2	4.9	4.4
L3	45.8	3.5	4.7	4.5
L4	88.3	0.8	4.6	4.4
L5	89.2	0.7	4.4	4.4
L6	88.4	1.2	4.3	4.1
L7	87.8	1.0	4.2	4.1
L8	84.9	1.6	4.3	4.1
L9	79.6	2.0	4.3	4.0
L10	66.7	3.3	4.2	4.1
L11	32.2	10.7	4.3	4.0
L12	7.5	5.0	4.3	4.0

Given two subspaces $\mathcal{A} \in \mathbb{R}^{d \times p}$ and $\mathcal{B} \in \mathbb{R}^{d \times q}$ (columns being the decoder directions), we first compute orthonormal bases $\mathbf{Q}_A$ and $\mathbf{Q}_B$ via QR decomposition, then obtain the principal angles $\theta_1, \dots, \theta_{\min(p,q)}$ via the SVD of $\mathbf{Q}_A^\top \mathbf{Q}_B$: the singular values are $\cos \theta_i$. We summarize subspace agreement as the mean of $\cos^2 \theta_i$ across all principal angles — a value of 1.0 indicates identical subspaces, while 0 indicates orthogonal subspaces.

As a baseline, we repeat the computation with two random subspaces of matching dimensionality (directions drawn uniformly from the unit sphere in $\mathbb{R}^{2048}$). As an upper bound, we compute PCA on the same activations and compare the top- k PCA subspace (matching the number of dense features) against each SAE’s dense subspace.

Results. The two SAEs’ dense subspaces overlap with a mean $\cos^2$ of 0.758. PCA on the same activations recovers the subspace with overlap 0.943 against either SAE. Both SAEs find the same geometric structure; the sparsity constraint forces them to express it in different, seed-dependent coordinates.

C Group Ablation of Dense Features

We test whether dense features are redundant — whether their high individual importance reflects genuine independent contributions or overlapping reconstructions of the same signal.

Method. We identify dense features in the reference SAE (seed 42, Pythia-1.4B, layer 12) as those whose decoder direction has $\cos^2$ with PC1 exceeding $5 \times$ the median across alive features, yielding 502 dense features. We compute two quantities:

Sum of individual importances. For each dense feature i , we set its activation to zero (leaving all other features intact), measure the increase in cross-entropy loss δ_i , and sum: $S = \sum_i \delta_i$.

Joint importance. We set all 502 dense features’ activations to zero simultaneously and measure the total loss increase Δ .

If the dense features encode independent information, $S \approx \Delta$. If they redundantly reconstruct the same signal, $S > \Delta$: each individual ablation causes a loss increase because the remaining $k-1$ active features cannot perfectly compensate, but the marginal effects overlap because all features are contributing to the same reconstruction.

Result. On Pythia-1.4B, $S/\Delta = 2.18$. The sum of individual importances is more than double the joint effect, confirming substantial redundancy. This is consistent with the non-identifiability described in Section 3.2: multiple dense features tile the same PC1 direction, so each one’s individual importance is inflated by measuring “how much does removing one piece of the tiling hurt” rather than “how much unique information does this feature carry.”

442 D Dense Feature Budget

Table 6: Dense features (decoder $\cos^2$ with PC1 $> 5 \times$ median) account for a small fraction of the alive dictionary but dominate total ablation importance.

Model	% of dictionary	% of importance
GPT-2-small	14.0	89.5
Qwen 2.5 7B	2.9	88.5
Pythia-160M	15.7	86.1
Pythia-1.4B	7.8	81.5
Pythia-6.9B	1.3	64.3
Llama 3.1 8B	8.1	53.6
Gemma 2 9B	3.5	51.8
Gemma 2 2B	8.9	49.0

443 E Training Details

444 Table 7 reports the target layer, hidden dimension, and dictionary size for each model. The target
445 layer is the middle layer of each model ($\lfloor L/2 \rfloor$ where L is the total number of transformer blocks).

Table 7: Model configurations and SAE training details. All SAEs use TopK with $k = 64$.

Model	Family	Layers	Target layer	d_{model}	d_{SAE}
Pythia-160M	Pythia	12	6	768	3,072
GPT-2-small	GPT-2	12	6	768	3,072
Pythia-1.4B	Pythia	24	12	2,048	8,192
Gemma 2 2B	Gemma	26	13	2,304	9,216
Qwen 2.5 7B	Qwen	28	14	3,584	14,336
Pythia-6.9B	Pythia	32	16	4,096	16,384
Llama 3.1 8B	Llama	32	16	4,096	16,384
Gemma 2 9B	Gemma	42	21	3,584	14,336
Llama 3.1 70B	Llama	80	40	8,192	32,768

446 All SAEs are trained with the following shared hyperparameters:

- 447 • **Training data:** 5M tokens from WikiText-103 (train split), tokenized with each model’s
448 native tokenizer, sequences truncated to 128 tokens.
- 449 • **Optimizer:** Adam with learning rate 3×10^{-4} , default $\beta_1 = 0.9$, $\beta_2 = 0.999$.
- 450 • **Batch size:** 4,096 activation vectors per step, sampled uniformly from the collected activa-
451 tions.
- 452 • **Steps:** 20,000.
- 453 • **Initialization:** Decoder columns initialized randomly and unit-normalized. Encoder initial-
454 ized as the transpose of the decoder. Biases initialized to zero.
- 455 • **Decoder normalization:** Decoder columns re-normalized to unit norm after each gradient
456 step.
- 457 • **Seeds:** 42, 137, 0, 1, 2 for all models except Llama 3.1 70B (seeds 42, 137 only).
- 458 • **Loss:** Mean squared error between the SAE’s output and the (centered) input activation. No
459 additional L1 penalty (sparsity is enforced by the TopK constraint).

460 For the PC1-bypass condition, we first collect all 5M activations, center them, compute PCA on a
461 50,000-activation subsample (float64), and subtract the projection onto the first eigenvector. The SAE
462 is then trained on these residual activations with the same hyperparameters. During evaluation, the
463 PC1 component is computed on-the-fly from the model’s hidden states and routed around the SAE.

464 Llama 3.1 70B is loaded in float16 with device_map="auto" across two H200 GPUs. All other
465 models fit on a single H100 GPU.

F Matching Threshold Robustness

Our main results use a cosine similarity threshold of 0.7 to classify features as shared or orphan, following Paulo & Belrose [2025]. To verify that our conclusions do not depend on this choice, we recompute top-50 recovery on Pythia-1.4B across thresholds from 0.5 to 0.95.

Table 8: Top-50 recovery on Pythia-1.4B at varying cosine thresholds. The inversion (top-50 below baseline before bypass, above baseline after) holds across all thresholds.

Threshold	0.50	0.60	0.70	0.80	0.90	0.95
Standard baseline (%)	84	74	62	48	28	12
Standard top-50 (%)	10	10	10	8	6	6
Bypass baseline (%)	83	72	58	44	25	11
Bypass top-50 (%)	100	94	90	80	64	48

The inversion (top-50 recovery below baseline before bypass, above baseline after) holds across all thresholds tested. The absolute recovery rates shift — higher thresholds produce lower absolute sharing — but the qualitative pattern and the ordering between conditions are unchanged.

G PC Sweep

We test whether removing additional principal components beyond PC1 further improves top-50 recovery. On Pythia-1.4B, we train SAEs after subtracting the top k principal components for $k \in \{0, 1, 2, 5, 10, 20, 50\}$ and measure top-50 sharing across two seeds.

Table 9: Top-50 sharing (%) as a function of the number of principal components removed before SAE training (Pythia-1.4B). The jump occurs entirely at $k=1$; further removal does not help.

PCs removed	0	1	2	5	10	20	50
Top-50 (%)	4	86	90	84	80	74	84

The confound is concentrated in PC1. Removing it raises top-50 sharing from 4% to 86%; removing additional components produces no systematic improvement, with sharing fluctuating between 74–90%. This is consistent with the theoretical account in Appendix L: PC1 alone captures the accumulated drift, while PC2 and beyond carry content-discriminative variance whose removal is neither helpful nor harmful.

H Post-Hoc Exclusion

PC1 bypass requires retraining the SAE. A lighter alternative is to train a standard SAE and exclude confounding features post-hoc by their decoder alignment with PC1. We identify dense features as those whose decoder direction has $\cos^2$ with PC1 exceeding a threshold, then recompute the importance ranking over the remaining features.

On Pythia-1.4B, this method is robust to the choice of threshold. At $3\times$, $5\times$, $10\times$, $15\times$, $20\times$, and $50\times$ the median $\cos^2$, the post-hoc top-50 sharing rate is 96% in every case, compared to 10% without exclusion and 90% with full bypass. The post-hoc method slightly outperforms bypass because it benefits from the SAE having already learned some content features alongside the dense ones; bypass forces the SAE to use its entire capacity on the content signal, which may produce a slightly different feature set.

I Multi-Layer Robustness

Our account predicts that the confound should appear wherever PC1 dominance is high. On Pythia-1.4B, the bell-shaped layer profile (Appendix A) shows PC1 dominance peaks in middle layers and drops near the embedding and output. We verify that the importance–reproducibility inversion tracks this profile.

The main experiments target layer 12 (the middle layer). The inversion is present at layer 12 and at neighboring layers where PC1 dominance remains high, and bypass corrects it in each case. At layers near the embedding or output where PC1 explains $<10\%$ of variance, the inversion is absent because dense features do not dominate the importance ranking.

J Cross-Domain Robustness

We test whether the effect depends on the training corpus. On Pythia-1.4B (layer 12), we train two SAEs on 5M tokens from OpenWebText (instead of WikiText-103) and repeat the analysis.

The residual stream geometry is nearly identical: PC1 explains 93.4% of variance on OpenWebText (vs. 96.4% on WikiText), and $\cos(\text{PC1}, \bar{\mathbf{x}}) = 0.93$. Without exclusion, the top-50 sharing rate is 2%, lower than on WikiText (10%). After excluding dense features by PC1 alignment ($\cos^2 > 5 \times$ median), top-50 sharing rises to 100%. The effect is robust across thresholds: at $3 \times$, $7 \times$, $10 \times$, and $20 \times$ median, top-50 sharing remains 100%.

K Top- k Robustness

We verify that the inversion is not an artifact of the specific choice of top-50. On Pythia-1.4B (OpenWebText, after PC1 exclusion at $5 \times$ median), we compute sharing rates at top- k for $k \in \{20, 50, 100, 200, 500\}$.

Table 10: Sharing rates (%) at varying top- k on Pythia-1.4B (OpenWebText, after PC1 exclusion). All values exceed the dictionary-wide baseline of 68%.

	Top-20	Top-50	Top-100	Top-200	Top-500
After PC1 exclusion (%)	100	100	98	97	92

Top- k sharing remains far above the baseline at all values of k tested. The effect is slightly weaker at top-500 (92% vs. 100% at top-50), reflecting the fact that less important features are less strongly constrained by model structure and therefore less consistently recovered.

L Theoretical Account: Why PC1 Dominates

Section 3.1 establishes empirically that PC1 explains 70–99.9% of residual stream variance and that this dominance is created by training. Here we sketch a theoretical account of why this concentration occurs.

The residual stream is additive: each layer reads the current state and writes back to it,

$$\mathbf{x}_L = \mathbf{x}_0 + \sum_{\ell=0}^{L-1} f_{\ell}(\mathbf{x}_{\ell}). \quad (2)$$

The three mechanisms described in Section 2 (gradient drift, attention contraction, and skip-connection compounding) all imply that each layer’s output has a consistent component along a shared direction $\mathbf{v}$, plus a content component orthogonal to it:

$$f_{\ell}(\mathbf{x}) = \underbrace{\alpha_{\ell}(\mathbf{x})}_{\text{drift}} \mathbf{v} + \underbrace{\mathbf{r}_{\ell}(\mathbf{x})}_{\text{content}}. \quad (3)$$

Two properties of this decomposition explain why PC1 captures most of the total variance and why the dominance is specifically rank-1.

Coherent drift versus dispersed content. Because every layer’s drift points along the same direction $\mathbf{v}$, the contributions add constructively: drift variance grows superlinearly with depth. Content contributions point in different directions at each layer (early layers encode syntax, later layers encode semantics), so they combine roughly as a random walk and total content variance grows only linearly with L . This gives $\sigma_{\alpha}^2 / \sigma_r^2 \sim O(L)$, and PC1 variance explained $\approx \sigma_{\alpha}^2 / (\sigma_{\alpha}^2 + \sigma_r^2)$ scales with depth. For $L = 24$ (Pythia-1.4B), this predicts PC1 $\approx 96\%$, consistent with the 96.4% we observe.

534 **Rank-1 rather than low-rank.** The dominance is rank-1 because content variance is dispersed
535 across $d-1$ dimensions. The spectral gap between PC1 and PC2 is:

$$\frac{\lambda_1}{\lambda_2} \approx (d-1) \frac{\sigma_\alpha^2}{\sigma_r^2}. \quad (4)$$

536 For $d = 2048$ and $\sigma_\alpha^2/\sigma_r^2 \sim 24$, this ratio is $\sim 50,000$: the eigenvalue cliff after PC1 is sharp.

537 **Caveats.** This account assumes that layerwise drift coefficients α_ℓ are positively correlated across
538 layers (for superlinear growth) and that content residuals are approximately isotropic across the $d-1$
539 orthogonal dimensions. We have not measured these quantities directly. The account is consistent
540 with the empirical observations in Section 3.1 and Appendix A, but other mechanisms could produce
541 similar patterns. The paper’s empirical conclusions do not depend on this theoretical model — they
542 rest on the observed PC1 dominance and the non-identifiability result in Section 3.2.

543 M Preprocessing, Architecture, and Metric Robustness

544 We test whether the importance–reproducibility inversion depends on preprocessing, importance
545 metric, matching method, or SAE architecture. We train TopK and BatchTopK SAEs on GPT-2-small
546 and Pythia-1.4B under four preprocessing conditions (raw centered, raw unit-norm, post-LayerNorm,
547 post-LN unit-norm), with and without PC1 bypass, using two seeds (42, 137). We evaluate importance
548 via both zero-ablation and mean-ablation, and compute cross-seed matching using max-cosine, mutual
549 nearest neighbor (MNN), and Hungarian one-to-one matching.

Table 11: Top-50 sharing (%) under varying preprocessing and PC1 bypass (TopK SAEs, zero-ablation importance). Three matching methods are reported: max cosine similarity (MaxCos), mutual nearest neighbor (MNN), and Hungarian bipartite matching (Hung).

Model	Preprocessing	Bypass	MaxCos	MNN	Hung	Dense % top-50
GPT-2	raw_centered	No	10	10	8	96
GPT-2	raw_centered	Yes	74	72	72	4
GPT-2	raw_unitnorm	No	36	34	34	56
GPT-2	raw_unitnorm	Yes	62	62	62	22
GPT-2	post_ln	No	74	70	74	34
GPT-2	post_ln	Yes	48	48	46	14
Pythia-1.4B	raw_centered	No	8	8	8	98
Pythia-1.4B	raw_centered	Yes	88	88	88	12
Pythia-1.4B	raw_unitnorm	No	4	2	4	100
Pythia-1.4B	raw_unitnorm	Yes	70	70	70	4
Pythia-1.4B	post_ln	No	68	66	68	30
Pythia-1.4B	post_ln	Yes	86	86	86	8

550 **Key findings.** Unit-norm normalization does not resolve the confound on Pythia-1.4B: top-50
551 sharing drops from 8% to 4%, and 100% of the top-50 features are dense. PC1 bypass raises sharing
552 to 70% regardless of normalization. On GPT-2, unit-norm partially mitigates the inversion (10% $\rightarrow$
553 36%) but PC1 bypass provides additional improvement. Post-LayerNorm activations show weaker
554 PC1 dominance and correspondingly less severe inversion, but PC1 bypass still helps on Pythia-1.4B
555 (68% $\rightarrow$ 86%).

556 Mean-ablation importance produces nearly identical results to zero-ablation: on Pythia-1.4B (raw
557 centered), top-50 sharing is 6% without bypass and 90% with. The inversion is not an artifact of the
558 zero-ablation metric.

559 All three matching methods (max cosine, MNN, Hungarian) agree to within 2 percentage points
560 throughout, confirming that many-to-one matching does not inflate our results.

561 BatchTopK SAEs show the same pattern as TopK on both models (2% $\rightarrow$ 92% on Pythia-1.4B raw
562 centered).

563 N Cross-Seed Steering Consistency

Table 12: Cross-seed decoder similarity predicts steering consistency. Negative Spearman ρ means higher decoder cosine gives lower steering KL, i.e. more consistent causal effects.

Model	Spearman ρ (cosine, KL)	p -value
Pythia-160M	-0.489	7.3×10^{-5}
Pythia-1.4B	-0.611	2.2×10^{-7}
Pythia-6.9B	-0.349	6.2×10^{-3}
GPT-2-small	-0.463	2.0×10^{-4}
Gemma 2 2B	-0.480	1.0×10^{-4}
Gemma 2 9B	+0.021	0.87
Llama 3.1 8B	-0.664	7.4×10^{-9}
Llama 3.1 70B	-0.552	4.8×10^{-6}
Qwen 2.5 7B	-0.631	6.4×10^{-8}

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction summarize the paper’s contributions: identifying an importance–reproducibility inversion for SAE features, attributing it to residual-stream anisotropy, proposing PC1 bypass, and evaluating the intervention across multiple model families with geometric and causal-consistency analyses. The experimental sections support these claims, especially the broad top-50 recovery improvements after PC1 bypass. Limitations are discussed in Section 7.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 discusses limitations including TopK/BatchTopK scope, English-only evaluation, causal validation concentrated on Pythia-1.4B, and open questions about what PC1 encodes.

3. Theory assumptions and proofs

Answer: [Yes]

Justification: The theoretical account given in Appendix L states its assumptions (positively correlated drift coefficients, approximately isotropic content residuals) and notes they have not been directly measured; the paper’s empirical conclusions do not depend on the theory.

4. Experimental result reproducibility

Answer: [Yes]

Justification: Section 4 and Appendix E specify all models, layers, hyperparameters, seeds, and evaluation procedures, and code is included as supplementary material.

5. Open access to data and code

Answer: [Yes]

Justification: Code for all experiments is included as supplementary material. All models and training data are publicly available.

6. Experimental setting/details

Answer: [Yes]

Justification: Section 4 describes models, SAE architecture, training procedure, PC1 estimation, and definition of key metrics used; Appendix E provides full hyperparameters.

7. Experiment statistical significance

Answer: [Yes]

600 Justification: Table 1 reports mean $\pm$ std across seed pairs for five-seed models, steering
601 consistency reports Spearman correlations with p -values (Appendix N), logit lens reports
602 Mann-Whitney U p -values.

603 **8. Experiments compute resources**

604 Answer: [Yes]

605 Justification: Appendix E reports GPU types (H100, H200), and that Llama 70B uses two
606 H200 GPUs while all other models fit on a single H100.

607 **9. Code of ethics**

608 Answer: [Yes]

609 Justification: The research conforms with the NeurIPS Code of Ethics. No human subjects,
610 no privacy concerns, no deceptive practices.

611 **10. Broader impacts**

612 Answer: [N/A]

613 Justification: This is interpretability research studying SAE feature reproducibility, and
614 as such, it does not introduce significantly new capabilities or models with direct paths to
615 negative applications.

616 **11. Safeguards**

617 Answer: [N/A]

618 Justification: The paper releases SAE analysis code, which does not have a high risk for
619 misuse.

620 **12. Licenses for existing assets**

621 Answer: [Yes]

622 Justification: All models used are publicly released under open licenses, WikiText-103 is
623 released under CC-BY-SA 3.0.

624 **13. New assets**

625 Answer: [Yes]

626 Justification: Code is released as supplementary material with documentation.

627 **14. Crowdsourcing and research with human subjects**

628 Answer: [N/A]

629 Justification: No crowdsourcing or human subjects research.

630 **15. Institutional review board (IRB) approvals or equivalent for research with human
631 subjects**

632 Answer: [N/A]

633 Justification: No human subjects research.

634 **16. Declaration of LLM usage**

635 Answer: [N/A]

636 Justification: LLMs are not used as an important, original, or non-standard methodological
637 component of this research.